1. Literature Review
The introduction establishes that suicide rates among medical students globally range from 3.8–18.7%, notably 3–5 times higher than the general population, driven by academic pressure, burnout, and limited institutional support. It reviews existing South Asian evidence gaps: Ahmed et al.'s WHO-SEA review found only 19 studies over 60 years, Rahman et al. found just 6 Bangladesh-specific studies over 24 years, and a broader South Asian scoping review found highly variable suicide rates (0.43–331 per 100,000) — establishing that no comprehensive meta-analysis has yet synthesized medical-student-specific suicidal behavior data for the region.

2. Methodology
This is a **study protocol** (not yet executed) for a systematic review and meta-analysis, following PRISMA-P and MOOSE guidelines, registered on PROSPERO. It will search seven databases (Medline, Scopus, Web of Science, EMBASE, PsycINFO, Google Scholar, regional databases) for studies (2000–2025) on suicidal ideation/attempts/planning among South Asian medical students, using validated tools (PHQ-9, C-SSRS, BDI, etc.). Screening will use EndNote and Rayyan with dual independent reviewers, quality assessed via a modified Newcastle-Ottawa Scale, and data pooled using random-effects meta-analysis in R.

3. Limitations
Anticipated limitations (stated proactively since this is a protocol): (a) expected high heterogeneity in study designs, assessment tools, and sample sizes across the region, which may challenge pooling and limit generalizability; (b) reliance on self-reported measures, introducing potential reporting bias; (c) some South Asian countries (e.g., Afghanistan, Bhutan, Maldives) likely lack sufficient published evidence, creating geographic coverage gaps. These will be addressed via subgroup/sensitivity analyses, imputation techniques for missing data where feasible, and transparent narrative synthesis where quantitative pooling isn't possible.

4. Method Used
A **systematic review and meta-analysis protocol** using random-effects models (inverse variance method) to pool prevalence estimates, with heterogeneity assessed via the I^2 statistic, publication bias assessed via funnel plots/Egger's/Begg's tests, and quality assessed via a modified Newcastle-Ottawa Scale — implemented in R (v4.3.2) with the meta and metafor packages.

Deep Dive

*Note: Since this is a **protocol** rather than a completed study, there are no results yet to critique in depth (points 5, 6, 8 apply to planned methodology rather than executed findings/algorithms in the way they did for the other papers). I've adapted accordingly.*

5. Advantages and Disadvantages of the Method (Random-Effects Meta-Analysis Protocol)

Advantages:

- Random-effects models are well-suited given the authors' own expectation of substantial cross-study heterogeneity (different countries, tools, sample sizes), since fixed-effect models would falsely assume one universal true prevalence.
- Pre-registration on PROSPERO and strict adherence to PRISMA-P/MOOSE/Cochrane Handbook guidelines reduces the risk of selective reporting or post-hoc changes to methodology — a strong safeguard against bias.
- Planned subgroup analyses (by gender, degree level, country, assessment tool, institution type) will help disentangle sources of heterogeneity rather than just reporting a single pooled number.
- Dual independent screening/extraction with Cohen's kappa monitoring and third/fourth-reviewer adjudication is a robust design against individual reviewer bias.

Disadvantages:

- Random-effects pooling across countries with vastly different cultural, economic, and healthcare contexts (Afghanistan to Sri Lanka) risks producing an average that doesn't meaningfully represent any single country's reality — a criticism seen in the "South Asia" depression meta-analysis previously reviewed, where India dominated the pooled sample.
- Reliance on self-reported prevalence data (inherited limitation of the underlying primary studies, not the meta-analytic method) means underreporting due to stigma or criminalization of suicide in some South Asian countries could systematically bias pooled estimates downward.
- The "no language restriction" inclusion criterion, while inclusive, introduces translation/interpretation challenges during screening and data extraction that could affect consistency.
- Since this is a protocol, all these strengths remain theoretical until execution — actual heterogeneity, publication bias, and study quality issues cannot be assessed until the review is completed.

6. How the Method Will Work (Detailed)

1. **Registration & framework**: The review is pre-registered on PROSPERO (CRD420251032718) and structured using the PECO framework (Population = South Asian medical students; Exposure = enrollment in medical school; Comparison = none, since it's a prevalence study; Outcome = suicidal ideation/attempt/planning).
2. **Search strategy**: A comprehensive search across seven databases plus regional repositories (BanglaJOL, NepJOL) will combine MeSH terms and free-text keywords for two core concepts — "suicidal behaviours" and "South Asian medical students" — using

Boolean operators, truncation, and database-specific filters, with no language or geographic-country restriction within South Asia.

3. **Screening**: References will be deduplicated in EndNote, then screened via Rayyan by two independent reviewers at both title/abstract and full-text stages, following the PRISMA 2020 flow diagram. Discrepancies go to a third reviewer; inter-rater reliability will be tracked via Cohen's kappa (≥0.60 required for acceptable agreement).
4. **Data extraction**: A structured extraction form captures study design, sample size, demographics, assessment tool, prevalence proportions (with 95% CIs), and risk factor estimates (ORs/RRs, converted to log odds ratios for consistency). Missing data will trigger author contact requests; unresolved gaps are either excluded from meta-analysis but retained in narrative synthesis, or imputed depending on the extent of missingness.
5. **Quality/bias assessment**: Each study will be scored (0–10) via a modified Newcastle-Ottawa Scale across three domains (sample representativeness/size, comparability, outcome assessment/statistical rigor), plus the GRADE approach for overall evidence certainty — with studies categorized as high/moderate/low risk of bias, though all will still be included in analysis (with sensitivity analysis excluding low-quality studies to test robustness).
6. **Statistical synthesis**: Pooled prevalence will be calculated separately for ideation, attempts (fatal vs. non-fatal), and planning using a random-effects model with the inverse variance method (weighting studies by precision), computed in R's meta package. Heterogeneity ($I^2 > 50\%$ = substantial) will trigger subgroup analyses (by gender, degree level, curriculum, country, institution type, assessment tool used).
7. **Publication bias**: Assessed via Begg's rank-correlation test, Egger's linear regression test for funnel plot asymmetry, and visual funnel plot inspection (if ≥10 studies per outcome), with a "trim-and-fill" procedure applied if asymmetry is detected, implemented via the metafor package.

7. Suitable Contexts/Problems

This type of protocol-driven, PRISMA-compliant meta-analytic methodology is best suited for:

- Synthesizing fragmented, geographically dispersed observational evidence on a stigmatized, underreported health outcome (suicidal behavior) where no single large study exists and small, heterogeneous studies must be aggregated to estimate population-level burden.
- Establishing baseline regional prevalence estimates to guide policy and resource allocation in under-studied regions (South Asia) where mental health research capacity and surveillance systems are limited.

- Situations requiring transparent, reproducible, and bias-minimized synthesis — the pre-registration and strict guideline adherence (Cochrane, PRISMA-P, MOOSE) are particularly valuable when findings will inform sensitive public health/policy decisions.
- It is less suited for real-time monitoring or capturing rapidly evolving trends, since systematic reviews are inherently retrospective and time-intensive (this one anticipates 15–18 months to complete) — a "snapshot in time" rather than a dynamic surveillance tool.

8. Anticipated Results/Findings

Since this is a **protocol paper**, no actual results exist yet — the paper does not report a completed meta-analysis. The authors do, however, provide preliminary contextual figures: globally, medical student suicidal ideation prevalence ranges from 3.8–18.7%, notably higher than the general population and university students broadly (illustrated via a heatmap/Sankey diagram compiled from other published sources, not from their own review). A preliminary PubMed search on April 12, 2025, retrieved 20 potential studies, but the full screening, extraction, and pooled analysis remain to be conducted. The protocol states an expected completion timeline of 15–18 months for the actual review and synthesis.

9. Overall Opinion and Critical Evaluation

As a protocol, this paper should be judged on the rigor and soundness of its planned methodology rather than on empirical findings, and on that basis it is a well-constructed, thorough plan. The explicit PECO framework, PROSPERO pre-registration, detailed PRISMA-P/MOOSE adherence, dual-reviewer screening with kappa-monitored agreement, and pre-specified subgroup/sensitivity/publication-bias analyses collectively represent strong methodological transparency — publishing a protocol before conducting the review is itself good practice, as it commits the authors to their stated analytic plan and reduces the risk of post-hoc "outcome switching" bias.

That said, a few points warrant scrutiny. First, the protocol's own introduction already flags that some South Asian countries (Afghanistan, Bhutan, Maldives) likely have little to no data — meaning the eventual "South Asia" pooled estimate risks being disproportionately India/Bangladesh/Pakistan-driven, similar to the pattern seen in the related depression-and-suicide South Asia review discussed earlier, without necessarily representing the full region as titled. Second, the reliance on self-reported prevalence data via validated instruments (PHQ-9, C-SSRS, etc.) is a structural limitation inherited from the primary literature that no meta-analytic technique can fully correct, especially given documented stigma and underreporting concerns the authors themselves cite. Third, the protocol's ambitious scope (distinguishing fatal vs. non-fatal attempts, ideation, planning, plus risk factor meta-analysis, plus multiple subgroup stratifications) is scientifically valuable but may prove difficult to execute if — as anticipated — few studies per subgroup exist, potentially forcing much of the analysis into narrative rather than quantitative synthesis, which the authors acknowledge as a contingency. Overall, this is a

methodologically sound and much-needed protocol; its ultimate value will depend on how many usable studies the completed search actually yields across the region's genuinely underserved countries.